

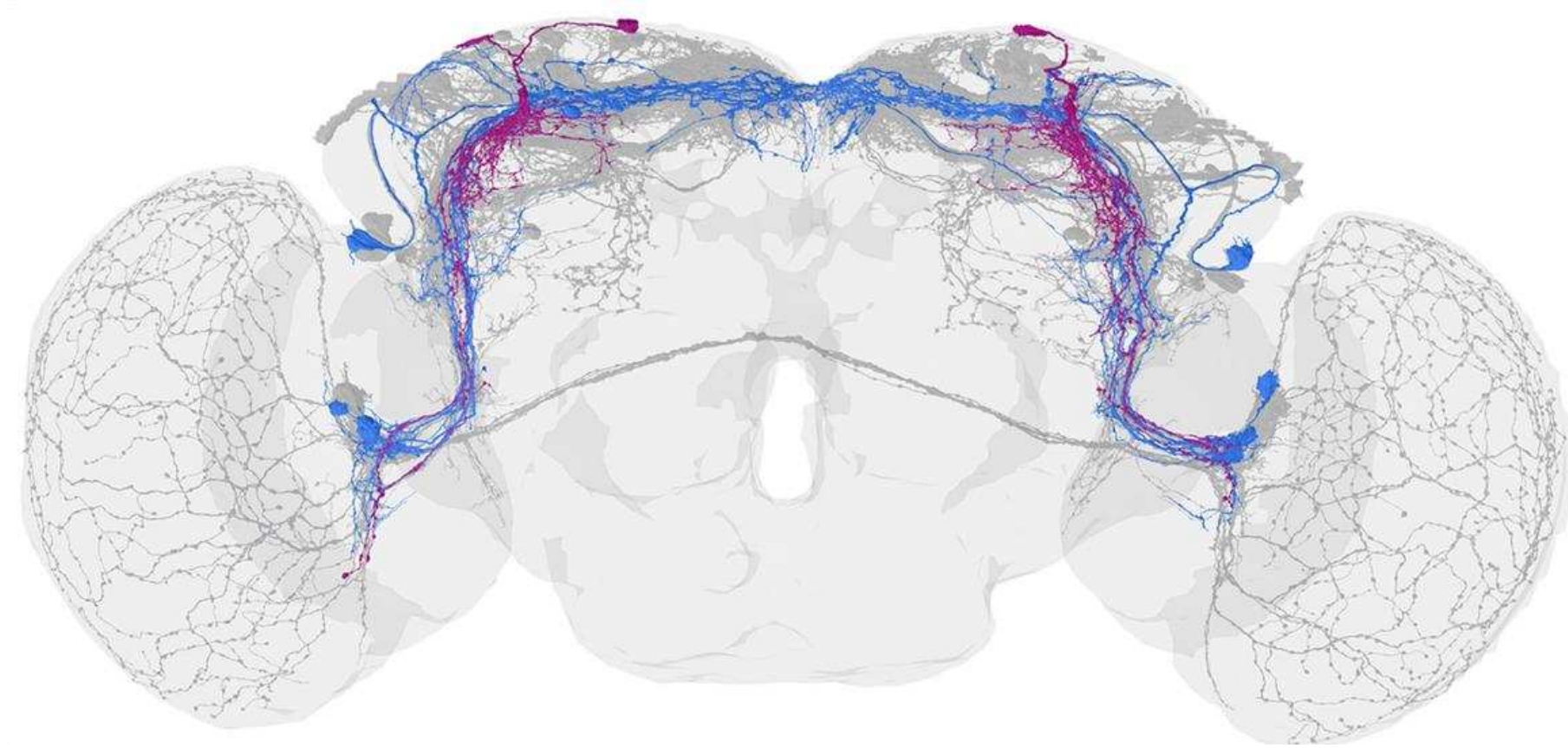


Figure 1: Fly circuit of the circadian clock (Universitat Würzburg, 2026)

Circadian Phase-Dependent Effects of Zeitgeber Time Disruption on Functional Mobility in *Drosophila melanogaster*



Figure 2: *Drosophila melanogaster* (Mullan & Marsh, 2019)

Introduction

- Circadian rhythms regulate key behavioral and physiological processes
- Circadian rhythms regulate sleep wake cycles, hormonal secretion, metabolism, and cognitive function (Desai et al., 2024)
- Modern lifestyle factors** such as shift work, prolonged light exposure to devices, and irregular sleep schedules **disrupt circadian rhythm** (Smith et al., 2025)
- Zeitgeber Time (ZT)** provides a standardized framework for studying circadian phase by **aligning internal biological clocks with external environmental cues**, primarily light (Northwestern University, 2022)
- Allows researchers to **introduce disruptions** at defined phases of the cycle
- Most research does not isolate specific ZT phases and focuses **on molecular responses** rather **than functional outcomes** (Lateef et al., 2023)

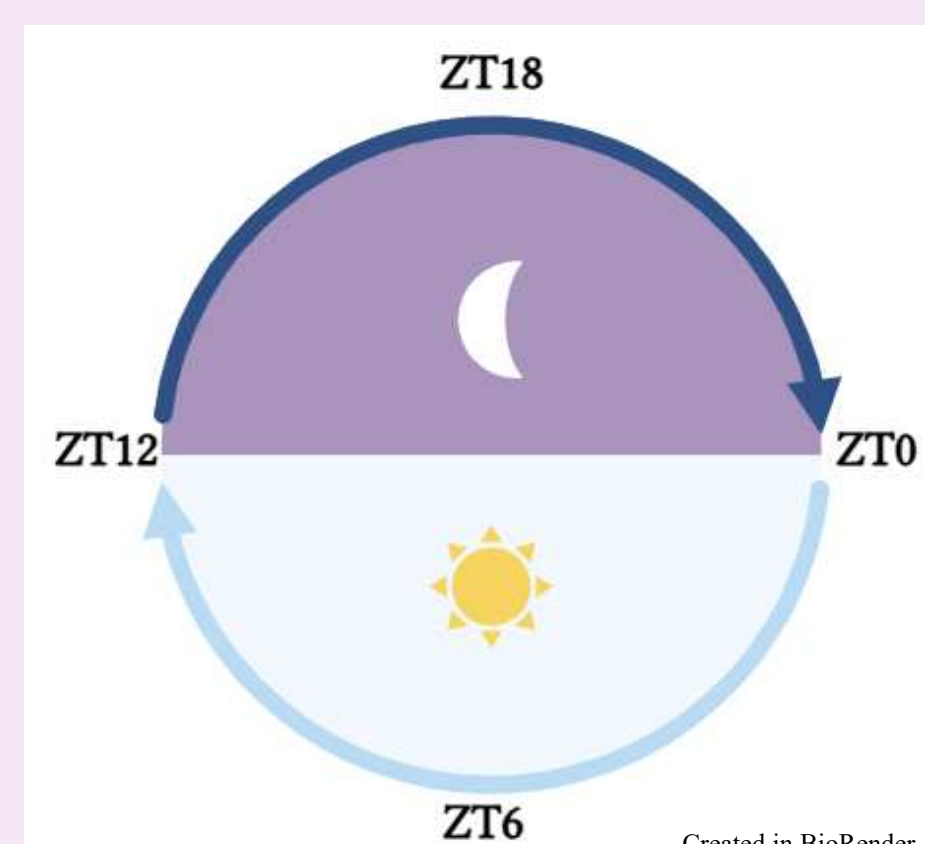


Figure 3: Zeitgeber time phases for flies

Methods

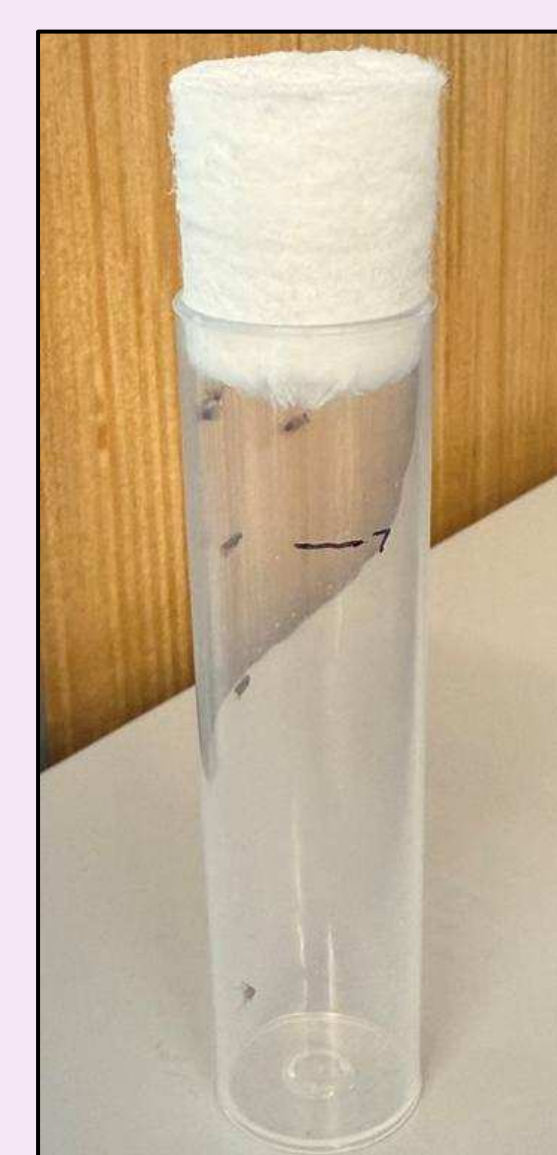
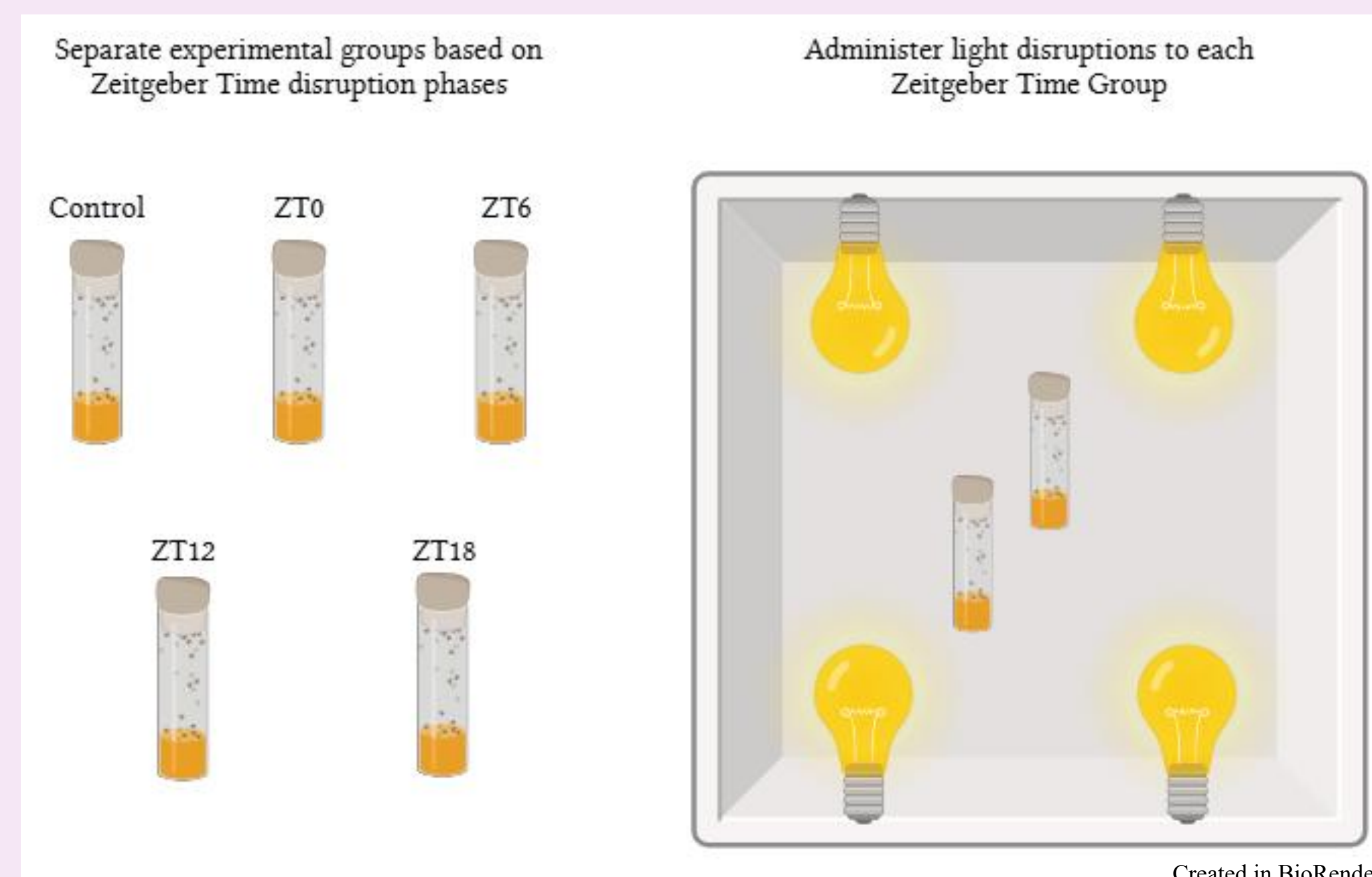


Figure 4: Experimental setup for negative geotaxis assay



Figure 5: Experimental setup for RING assay

Experimental Procedure

- Train flies on a 12:12 light dark cycle for 5 days at low intensity light
- Divide flies into experimental groups
- Administer light disruptions for each respective group for 5 days at a high intensity light
- Assess functional mobility with negative geotaxis and RING assay
- Conduct statistical analysis using an ANOVA test

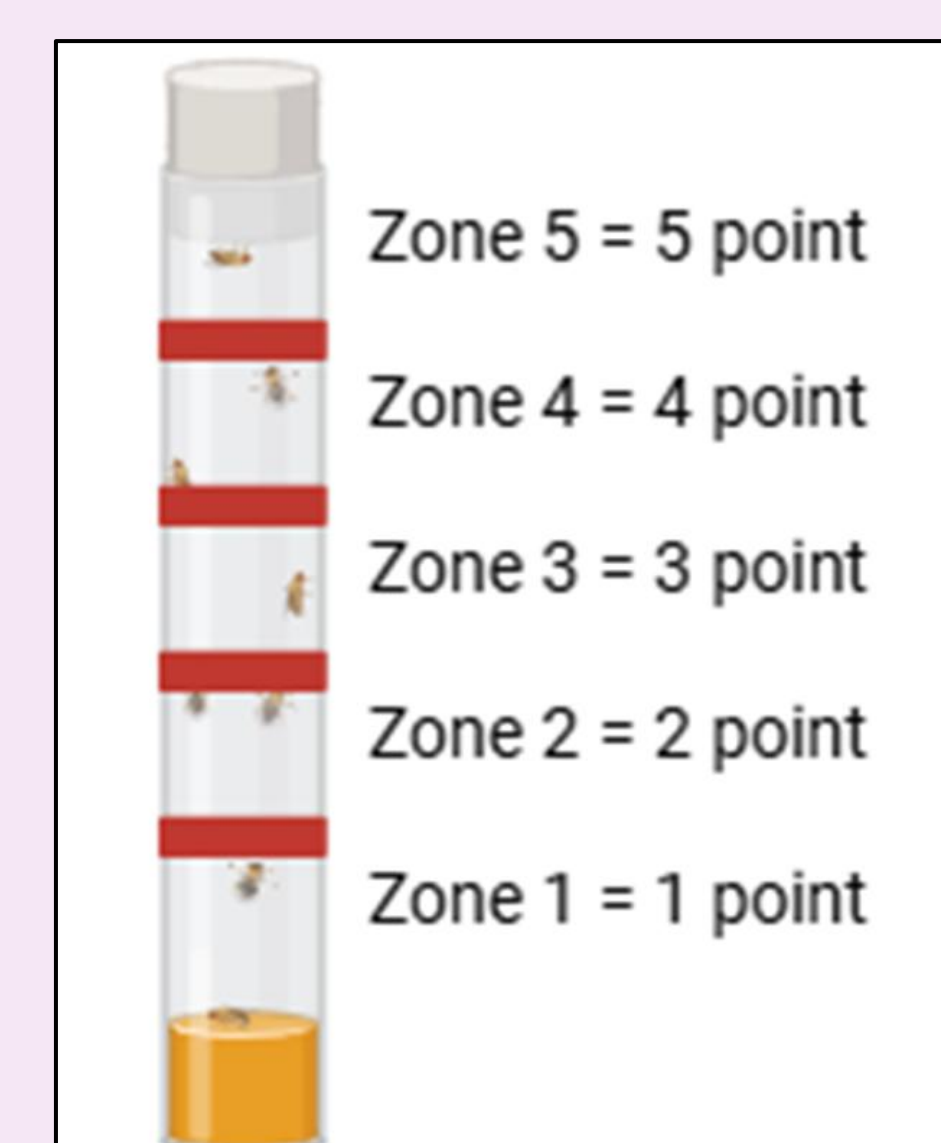


Figure 6: Weighted climbing score methods

Conclusion

- Results show that timing of circadian disruption has a statistically significant impact on functional mobility in *Drosophila melanogaster*
- Across both behavioral assays, there was extremely strong evidence that at least one experimental group differed from the others
- In the negative geotaxis assay, flies exposed to **ZT12 disruption showed a highly significant decrease in mobility** compared to the control group ($p < 0.01$)
- Similarly, comparisons between **ZT12 and other groups (ZT0 and ZT6)** also yielded statistically significant differences ($p < 0.01$)
- Reinforces that ZT12 had the most pronounced effect
- In contrast, **ZT0 ($p=0.80$) and ZT6 ($p=0.87$)** showed **no statistically significant difference** from the control, while **ZT18 ($p=0.19$)** suggested only a **non significant trend** towards reduced mobility
- In conclusion, this study highlights the importance of circadian phase in determining the **severity of behavioral and physiological disruptions**
- The findings suggest that not all circadian disturbances are equally impactful, and that disruptions during key transition periods may have **disproportionately large effects**
- Future research can work on investigating these effects in terms of molecular responses

Objectives

- Purpose:** Investigate how disruptions at different Zeitgeber times affect functional mobility in flies.
- Hypothesis:** Flies exposed to disruption at ZT12 would have a greater decline in functional mobility because clock proteins are most sensitive here.
- Independent Variable:** Zeitgeber time disruption phase (ZT0, ZT6, ZT12, ZT18)
- Dependent Variable:** Motor functions
- Controls:** Group with no light disruption
- Constants:** Fly sex, gender (male), age, diet, temperature, time of data collection

Results

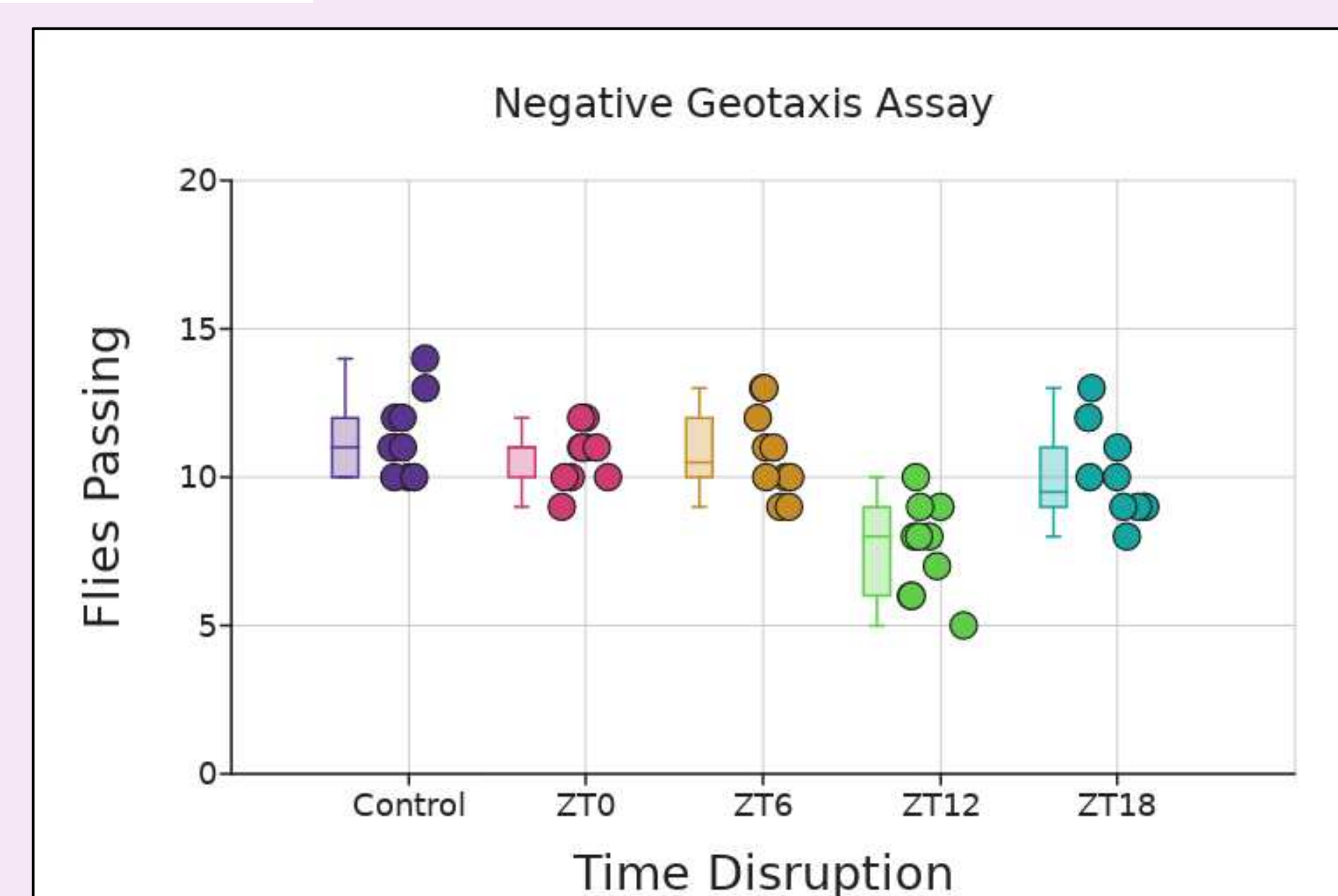


Figure 7: Negative geotaxis assay graph

Key Takeaway: The number of flies passing is lower for the ZT12 group

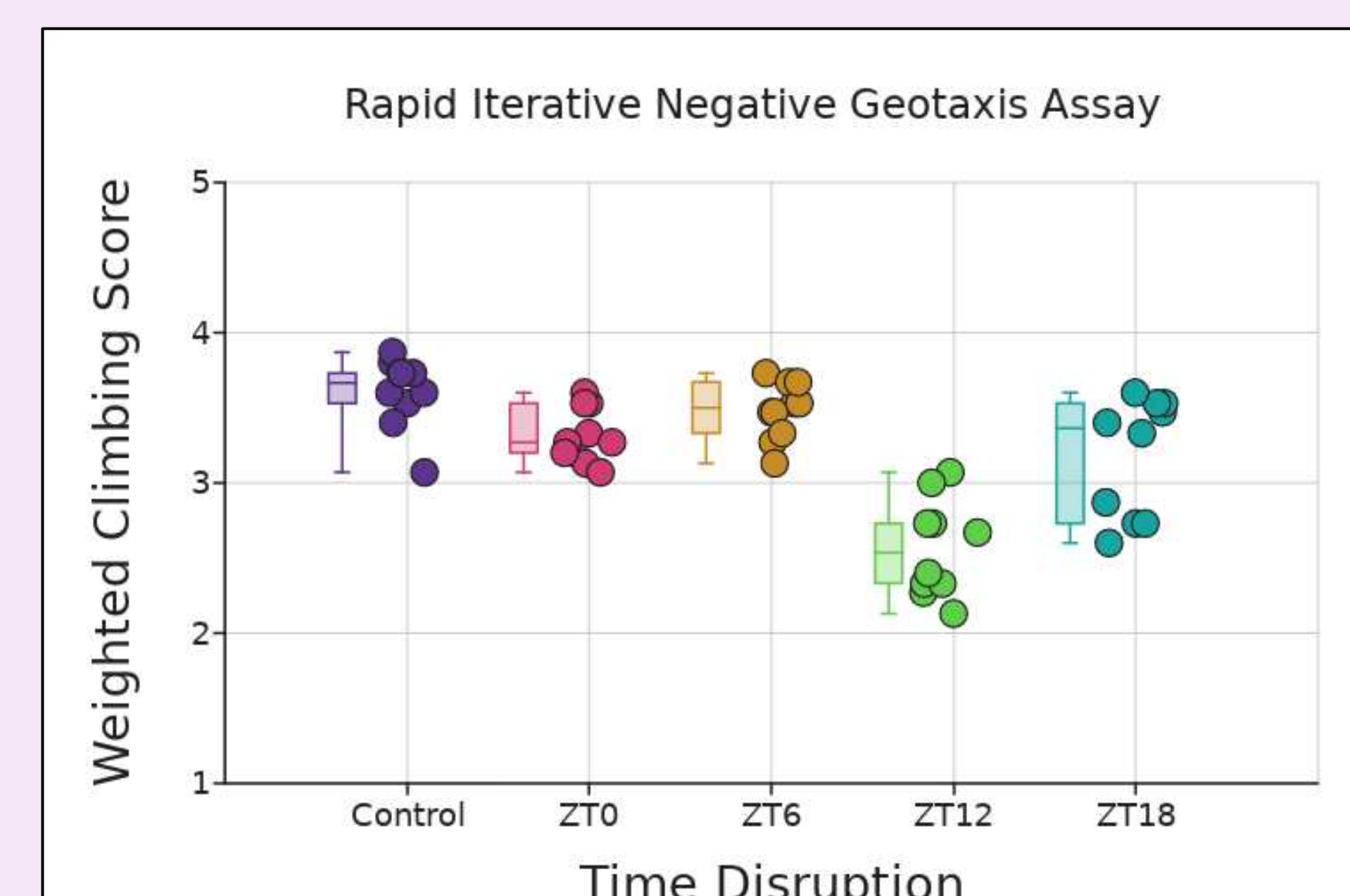


Figure 8: RING assay graph

Key Takeaway: The average weighted climbing score is lower for the ZT12 group

References

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